Traffic CO2 Emission Monitor

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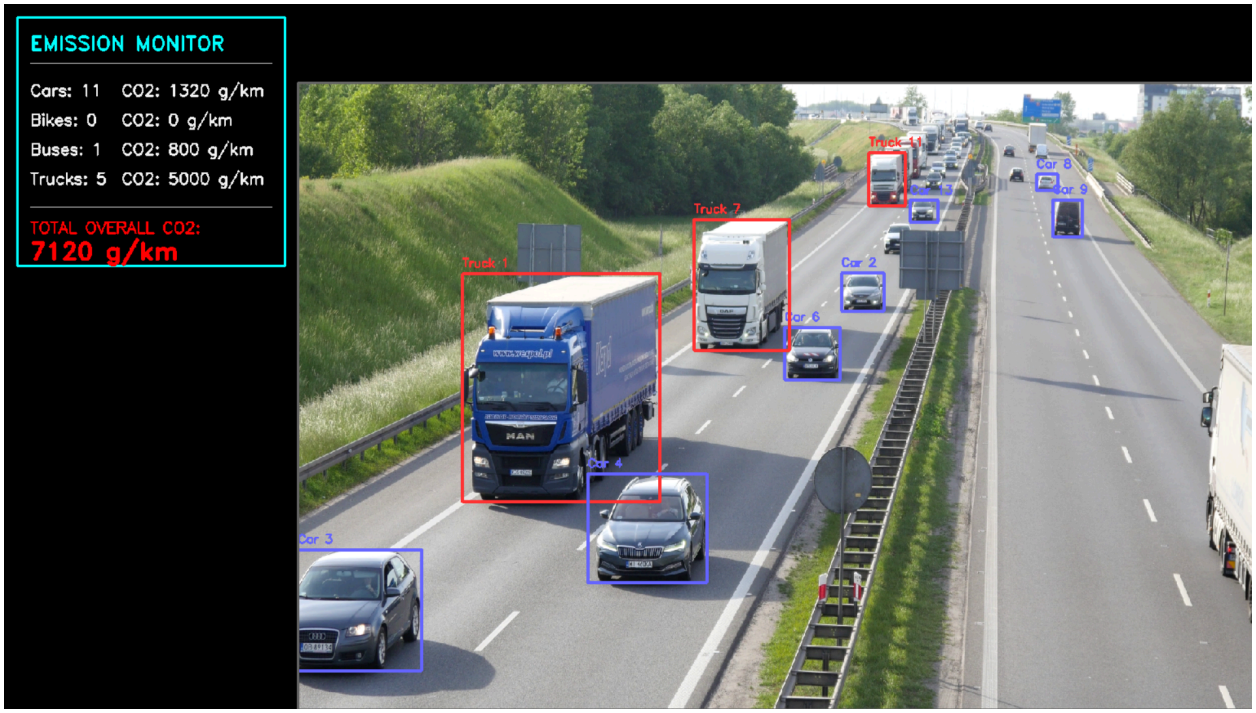
Project Overview

This application leverages computer vision to monitor traffic flow and estimate carbon dioxide (CO2) emissions in real-time. By utilizing object detection to identify different vehicle types, it provides an immediate visual dashboard of environmental impact.

Live Tracking Dashboard



The live dashboard displaying real-time bounding boxes, vehicle classifications, and the running CO2 emission tally.

1. Core Features

- **Real-Time Object Tracking:** Uses a pre-trained YOLO model to detect, classify, and track unique vehicles across video frames.

- **Dynamic Emission Calculation:** Calculates cumulative CO2 emissions (in g/km) based on predefined emission factors for different vehicle classes.
- **Custom Dashboard UI:** Constructs a unique picture-in-picture style layout entirely using NumPy and OpenCV array slicing, avoiding the overhead of heavy GUI frameworks.
- **Automated Reporting:** Generates a formatted summary report in the terminal upon closing the application.

2. Technical Stack

Component	Technology / Library	Purpose
Machine Learning	Ultralytics YOLO (yolov8n.pt)	High-speed vehicle detection and tracking.
Image Processing	OpenCV (cv2), NumPy	Video decoding, frame resizing, bounding box rendering, and canvas creation.
File Handling	Tkinter	Provides a native OS dialog for easy video file selection.

3. Vehicle & Emission Metrics

The system tracks four distinct COCO classes. Each vehicle is assigned a specific emission rate and a unique color for the bounding boxes in the video feed:

Vehicle Class	CO2 Emission Rate (g/km)	Bounding Box Color
Bike	80	Light Green
Car	120	Light Blue

Bus	800	Yellow
Truck	1000	Red

4. User Workflow & Final Reporting

1. **Initialization:** The script opens a standard file explorer dialog prompting the user to select a video file (.mp4, .avi, etc.).
2. **Execution:** The full-screen UI launches, playing the video with superimposed tracking boxes and updating the emission dashboard in real-time.
3. **Termination:** The user presses the q key to safely exit the video loop.
4. **Summary:** The application cleans up its windows and prints a formatted summary report directly to the command-line terminal.

Terminal Summary Output

```
=========================================================
            FINAL TRAFFIC & EMISSION REPORT
=========================================================
Vehicle Type    | Total Count     | Total CO2 (g/km)
---------------------------------------------------------
Cars            | 188             | 22560
Bikes           | 363             | 29040
Buses           | 46              | 36800
Trucks          | 105             | 105000
---------------------------------------------------------
OVERALL TOTAL   | 702             | 193400
=========================================================
```

The final printed summary detailing total counts and CO2 contributions per vehicle category, alongside the overall environmental impact.

5. Conclusion

The Traffic CO2 Emission Monitor provides a highly visual, practical solution for assessing the environmental impact of vehicular traffic. By combining the speed of YOLOv8 object detection with a custom, lightweight OpenCV interface, this tool offers immediate insights into carbon emissions without requiring heavy external dependencies. It serves as an excellent foundation for broader smart-city applications, traffic management studies, or localized environmental monitoring initiatives.